Applying Sleep-Dependent Memory Consolidation to Agent Architecture

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Abstract

Current LLM agent memory systems rely on context window accumulation, lossy summarization, or explicit user management—none of which produce improved signal-to-noise ratio over time through selective forgetting. We present a memory architecture for autonomous AI agents inspired by sleep-dependent memory consolidation in biological neural systems. Drawing on Complementary Learning Systems theory, synaptic homeostasis, and memory reconsolidation, we implement a two-store model with offline consolidation: a fast-encoding episodic store for real-time context ingestion and a slow-learning structured store for consolidated long-term memory, connected by a scheduled "dreaming" pipeline. The system extracts five categories of structured memory from raw multi-service context, performs two-stage matching with contradiction detection and reconsolidation, applies power-law trace-strength decay, and logs training signals for transitioning from heuristic to learned consolidation. We describe the architecture, a seven-primitive cortex abstraction layer, embedding infrastructure, and training pipeline. The system is implemented in ∼3,200 lines of Go on PostgreSQL and deployed to production. We position the work against recent developments including FadeMem, Phasor Agents, SleepGate, Agentic Memory, and AriadneMem, and discuss limitations including the absence of benchmark evaluation and LLM-dependent consolidation quality.

1 Introduction

Every agent framework in production today handles memory through one of three mechanisms: context window accumulation (bounded by token limits, $O(n)$ degradation in retrieval quality), lossy summarization (irreversible information destruction with no quality signal), or explicit user-managed memory (shifting cognitive load onto the human). None of these produce the outcome that biological memory systems achieve: improved signal-to-noise ratio over time through selective forgetting.

The neuroscience is unambiguous on this. Tononi and Cirelli's synaptic homeostasis hypothesis [Tononi and Cirelli, 2003, 2014] demonstrates that waking experience potentiates synaptic connections broadly, while sleep globally downscales synaptic strength. Connections reinforced by repeated activation are selectively preserved. The net effect is that an organism wakes with fewer total synaptic connections but higher fidelity recall of information that matters. Memory improves not despite forgetting, but because of it.

McClelland, McNaughton, and O'Reilly's Complementary Learning Systems (CLS) theory [McClelland et al., 1995], updated by Kumaran et al. [2016], provides the architectural framework. Intelligent memory requires two interacting systems: a fast-learning episodic store (the hippocampus) that rapidly encodes new experiences, and a slow-learning structured store (the neocortex) that gradually integrates patterns across experiences. The critical transfer

mechanism is offline consolidation during sleep, where hippocampal replay drives neocortical integration.

Rasch and Born [2013] showed that this consolidation is not passive. It is mediated by active replay of neural activation patterns during slow-wave sleep, selectively biased toward information tagged as relevant during encoding. The selection pressure is driven by emotional salience, repetition frequency, and congruence with existing knowledge schemas. Tse et al. [2007] demonstrated that schema-congruent information consolidates an order of magnitude faster than schema-incongruent information.

This paper describes the translation of these biological mechanisms into a production memory architecture for an AI agent platform. The architecture arrived during a 50-mile ultramarathon—endurance running suppresses prefrontal executive function and activates the same default mode network implicated in memory consolidation and creative problem-solving [Raichle et al., 2001]. This is noted not as anecdote but as context: the design is a direct product of the neuroscience it implements.

2 Architecture: A Two-Store Model with Offline Consolidation

Eva is an AI executive assistant platform. It connects to a user's Gmail, Calendar, Google Docs, GitHub, OneDrive, and Dropbox, syncing context every 30 minutes into an episodic store (`user_contexts`). This is the hippocampal analogue: fast-encoding, high-fidelity, time-bounded records of what happened.

The problem faced was identical to the CLS problem: the episodic store grows linearly with time, retrieval degrades (ILIKE substring matching on summary text), and hard time-based deletion at 7 days destroys information indiscriminately. There was no neocortical analogue—no structured long-term memory and no consolidation mechanism.

2.1 The Dreaming Pipeline

Every 6 hours, a scheduler job initiates a *dream session* for each user. The pipeline has five phases.

Phase 1: Gather. Query `user_contexts` for all records since the last completed dream session, ordered by encoding-time salience. Each context carries a salience score assigned at ingestion based on structural signals from its source: upcoming calendar deadlines, response-needed emails, failing CI builds, new collaborators. High-salience contexts are processed first, ensuring that if the token budget is exhausted mid-session, the low-importance contexts are the ones skipped. This mirrors the amygdala-hippocampus interaction where emotionally salient experiences are preferentially replayed during sleep.

Phase 2: Consolidate. Raw contexts are batched (n=20 per batch) and processed through an LLM with a structured extraction prompt. The prompt instructs the model to identify five categories of consolidated memory (Table 1).

Each extracted memory is assigned a confidence score ($c \in [0, 1]$) based on evidence density: 0.3–0.5 for single data points, 0.5–0.7 for multiple supporting observations, 0.7–0.9 for strong recurring patterns, 0.9–1.0 for explicit statements with corroborating behavioral evidence. Source context IDs are preserved for provenance tracing.

Phase 3: Match and Merge. Each LLM-produced memory undergoes two-stage matching against the existing neocortical store (`user_memories`).

Stage 1—Candidate generation: Jaccard similarity on tokenized summaries within the same category (threshold ≥ 0.2) generates candidate matches.

Table 1: Memory categories with biological analogues and selection criteria.

Category	Biological Analogue	Selection Criterion
`behavioral_pattern`	Procedural/implicit memory	Repeated action sequences with contextual triggers
`preference`	Evaluative conditioning	Consistent choice patterns or explicit preferences
`relationship`	Social cognition / person memory	Recurring entity co-occurrence with role differentiation
`project_context`	Episodic future thinking	Goal-directed activity with temporal trajectory
`routine`	Circadian/habitual memory	Temporally regular patterns with low variance

Stage 2—LLM confirmation with contradiction detection: Candidates are sent to the LLM for semantic classification. For each pair, the LLM determines whether the new memory reinforces the existing one (same insight, more evidence), contradicts it (opposite insight with evidence), or represents something genuinely new.

This two-stage approach eliminates both false positive reinforcement (high word overlap but different meaning) and false negative duplication (different wording but same semantic content).

For reinforcement: existing memory confidence is incremented (+0.05), reinforcement count is incremented, and `last_reinforced` timestamp is updated. This is the long-term potentiation analogue.

For contradictions: the system implements a reconsolidation protocol inspired by Nader et al. [2000]. The old memory is destabilized (confidence decayed at 2× rate, `contradicted_by` field set to the replacement's ID), the new memory is inserted, and a `contradicts` edge is created in the knowledge graph. This mirrors biological reconsolidation where new conflicting information destabilizes an existing trace, creating a window for update. Recent work by Cohen et al. [2025] validates this choice—they demonstrate that LLMs catastrophically destroy unrelated knowledge when trained on contradictory information, but explicit temporal contextualization (creating new memories rather than overwriting) preserves knowledge integrity.

Phase 4: Decay. All memories not reinforced during the current session undergo confidence reduction following a power-law forgetting curve [Wixted, 2004]:

$$\text{decay_rate} = \frac{\text{base_factor}}{1 + \ln(\max(\text{reinforced_count}, 1))} \tag{1}$$

Table 2: Decay rates as a function of reinforcement count.

Reinforced Count	Decay per Session	Sessions to Expiry (from 0.8)
1 (new)	0.150	∼5
5	0.063	∼12
20	0.038	∼20
100	0.024	∼32

A memory reinforced 20 times over months becomes nearly permanent. A weakly reinforced memory decays at the original rate. This models the biological finding that forgetting rate is inversely proportional to trace strength—strongly consolidated memories are resistant to interference, while weakly encoded traces are rapidly displaced.

Memories whose confidence falls below 0.1 receive an expiry timestamp (now + 7 days), set once and never extended. A memory that crosses the decay threshold is on an irreversible path

to deletion.

Phase 5: Record. A `dream_sessions` audit row captures contexts processed, memories created, memories reinforced, memories decayed, tokens consumed, and completion status. Every reinforcement, contradiction, and decay event is logged to `memory_training_events` for the learned scoring model (Section 4).

2.2 Concurrency

Dream sessions use an atomic check-and-insert guard:

```
INSERT INTO dream_sessions (user_id, status)
SELECT $1, 'running'
WHERE NOT EXISTS (
    SELECT 1 FROM dream_sessions WHERE user_id = $1 AND status = 'running'
)
RETURNING id
```

This eliminates TOCTOU races between concurrent triggers without advisory locks. Session completion uses a detached context ensuring the audit record is written even on parent context cancellation.

3 The Cortex Abstraction

The consolidation pipeline is wrapped in a higher-level abstraction called the *cortex*, exposing seven primitives (Table 3).

Table 3: Cortex primitives with biological analogues.

Primitive	Operation	Biological Analogue
Remember	Ingest observation with auto-embedding and decay scheduling	Hippocampal encoding
Consolidate	Run dream session: gather, extract, merge, decay	Sleep-dependent consolidation
Recall	Hybrid retrieval: salience + recency + semantic	Cue-dependent retrieval
Reinforce	Strengthen memory on successful use	Long-term potentiation
Decay	Power-law confidence reduction of unreinforced memories	Synaptic homeostasis
Relate	Create typed edge between memories in knowledge graph	Associative linking
Introspect	Structured summary of system knowledge about a user	Metacognition

The Recall scoring function combines two components, with an optional third:

$$\begin{aligned} \text{score}(q, m) = {} & 0.6 \cdot \text{salience}(m.\text{confidence}) \\ & + 0.4 \cdot \text{recency}\left(e^{-d/30}\right) \\ & [+ \text{semantic}(\cos(e_q, e_m))] \end{aligned} \tag{2}$$

where d is days since last reinforcement, e_q and e_m are query and memory embeddings respectively. Salience is the reinforcement-adjusted confidence value. The recency component applies exponential decay with a 30-day time constant, modeling the Ebbinghaus forgetting

curve. When embedding infrastructure is active, the semantic similarity term is blended in and the weights are rebalanced across all three components.

The INTROSPECT primitive renders the top-K highest-confidence memories per category as "What Eva Knows About You" in the agent's system prompt. This is primed recall: the most salient knowledge is immediately accessible without deliberate retrieval.

4 Training Infrastructure: From Heuristic to Learned Consolidation

The current pipeline uses LLM-generated confidence scores as the selection pressure. This is effective but not defensible. The architecture is designed to transition to learned consolidation. Recent work validates this direction: AgeMem [Xu et al., 2026] and Memory-R1 [Yan et al., 2025] demonstrate that RL-trained memory policies consistently outperform heuristic baselines across multiple benchmarks, with Memory-R1 achieving strong generalization from just 152 training pairs.

Every interaction generates training signals (Table 4).

Table 4: Training signals generated during agent interaction.

Event	**Signal**	**Source**
Memory retrieved by agent	`retrieved` (1.0)	`search_context` tool call
Memory used in successful wish	`led_to_success` (1.0)	Wish completion trace
Memory returned but not used	`returned_not_used` (-0.3)	Agent response analysis
Memory reinforced during dreaming	`reinforced` (1.0)	Dream merge phase
Memory decayed during dreaming	`decayed` (decay rate)	Dream decay phase

The inclusion of implicit negative signals—memories returned by search but not referenced in the agent's response—addresses survivorship bias. Without negatives, the model learns to replicate LLM confidence rather than predict genuine utility.

The target is a 2-layer MLP ($\sim$50K parameters) trained on composite labels weighted toward retrieval and success signals. Two-phase training: bootstrap on survival labels, then fine-tune on retrieval/success pairs. Cold start managed by LLM fallback until 5,000+ events accumulate ($\sim$2–4 weeks with 10 active users).

5 Related Work

The agent memory landscape has moved rapidly. A survey of the mid-2025 through early-2026 literature reveals several relevant developments, clustering around three themes: biologically-inspired consolidation mechanisms, learned memory management, and conflict detection with structured retrieval.

5.1 Consolidation Mechanisms

The consolidation work is the most directly relevant to our architecture. **FadeMem** [Wang et al., 2026] implements adaptive exponential decay modulated by semantic relevance, access frequency, and temporal patterns, achieving 45% storage reduction while improving multi-hop reasoning over Mem0. Our decay mechanism uses power-law trace-strength modulation rather than their exponential relevance-weighted approach. Theirs is stronger at pruning context-irrelevant memories; ours is stronger at preserving deeply reinforced knowledge. Neither system ingests multi-service life context, and FadeMem has no contradiction detection or offline consolidation.

Phasor Agents [Phasor Agents, 2026] provide the strongest empirical evidence that wake/sleep separation improves learning—67% expansion of stable learning under matched compute budgets, and REM-like replay improving performance by 45.5 percentage points. Our architecture implements wake/sleep separation (context sync during wake, dream sessions during sleep) with encoding-time salience tagging that mirrors their three-factor plasticity wake tagging.

SleepGate [SleepGate, 2026] implements sleep-inspired consolidation at the transformer level, augmenting the KV cache with conflict-aware temporal tagging, a forgetting gate, and a consolidation module that merges related entries into compact summaries. They reduce proactive interference from $O(n)$ to $O(\log n)$, achieving 99.5% retrieval accuracy at interference depth 5 versus <18% for all baselines including sliding window, H2O, and StreamingLLM. This is the strongest empirical validation that sleep-like consolidation works in neural architectures. Our system operates at a higher abstraction—whole-memory consolidation across heterogeneous life data rather than KV cache management—but the results are complementary: SleepGate demonstrates the mechanism is sound at the attention layer; we demonstrate it scales to application-level memory over multi-service context.

CLS-M [CLS-M, 2026] is the closest architectural cousin. It explicitly maps CLS theory to LLM infrastructure: dual-store memory with fast episodic capture and slow semantic consolidation, spreading activation over knowledge graphs, and decay-based forgetting. Across 200+ agent sessions it achieved 98% query coverage while reducing context token consumption by 96.5%. The parallel evolution validates the biological blueprint as a convergent design. Our differentiators are multi-service life context ingestion (not chat-only), the contradiction detection and reconsolidation protocol, encoding-time salience scoring, and the training infrastructure for transitioning to learned consolidation.

5.2 Learned Memory Policies

On the learned policy front, two systems validate the transition from heuristic to trained memory management. **Agentic Memory (AgeMem)** [Xu et al., 2026] exposes memory operations—store, retrieve, update, summarize, discard—as tool-based actions and trains unified memory policies through three-stage progressive RL (step-wise GRPO). **Memory-R1** [Yan et al., 2025] uses a two-agent architecture: a Memory Manager learning ADD/UPDATE/DELETE/NOOP operations via PPO/GRPO, trained on just 152 QA pairs with strong generalization across LoCoMo, MSC, and LongMemEval. Both outperform heuristic baselines, confirming that our planned transition from LLM-generated confidence scores to a learned scoring model is the right trajectory. Our approach differs in the training signal: they optimize for downstream task accuracy; we accumulate implicit signals from actual agent behavior (retrieval, use, ignore) to learn what mattered in practice rather than what a benchmark rewards.

5.3 Conflict Detection and Structured Retrieval

For conflict detection and structured retrieval, three systems address problems we encounter in Phase 3 and the knowledge graph. **The "Stubbornness" paper** [Cohen et al., 2025] validates our contradiction handling: contradictions are detectable with 95%+ accuracy and temporal contextualization (our approach: insert new, mark old) preserves unrelated knowledge. We are evaluating replacing the LLM contradiction call with a lightweight NLI classifier (`cross-encoder/nli-deberta-v3-small`, 22M parameters) that could achieve comparable accuracy at 200× lower cost.

AriadneMem [AriadneMem, 2026] addresses the same conflict detection problem from a different angle. Their "conflict-aware coarsening" merges static duplicates while preserving state transitions, improving Multi-Hop F1 by 15.2% on LoCoMo with 77.8% runtime reduction. Our two-stage approach (Jaccard candidate generation → LLM semantic classification) with explicit

contradiction edges provides richer provenance, but AriadneMem's coarsening strategy could improve merge efficiency for high-volume consolidation batches.

MAGMA [MAGMA, 2026] represents each memory across orthogonal semantic, temporal, causal, and entity graphs, decoupling memory representation from retrieval logic. This multi-graph decomposition addresses the causally grounded retrieval gap identified in the Du survey [Du et al., 2026]. Our RELATE primitive creates typed edges in a single knowledge graph; MAGMA's orthogonal decomposition suggests the graph should be stratified by relationship type to enable query-adaptive traversal.

5.4 Surveys and Evaluation

Two works frame the broader landscape. **The Du survey** [Du et al., 2026] identifies five open challenges for agent memory. We address continual consolidation and learned forgetting. The remaining three—causally grounded retrieval, trustworthy reflection, and multimodal embodied memory—are opportunities for future work. **MemoryAgentBench** [MemoryAgentBench, 2026] establishes an evaluation framework assessing four core memory competencies: accurate retrieval, test-time learning, long-range understanding, and selective forgetting. Current methods—from simple RAG to advanced agents with external memory modules—fall short across all four. The selective forgetting competency provides an external benchmark for validating our decay and contradiction mechanisms against the field.

6 Embedding Infrastructure

Memory tables carry optional `vector(768)` columns indexed with HNSW (m=16, ef_construction=64). HNSW was chosen over IVFFlat because it works incrementally on empty tables.

The embedding model recommendation is `nomic-ai/modernbert-embed-base`: 768 dimensions, 149M parameters, 8192 token context, Matryoshka representation learning (truncatable to 256 dimensions with ∼2% quality degradation), 62.62 MTEB average. Domain-specific evaluation against production memory summaries is planned before final model selection.

The entire embedding pipeline is a progressive enhancement. The same seven cortex primitives work with or without the embedding sidecar.

7 Implementation

The system is implemented across four packages:

- `internal/dreaming/`—Dream pipeline: gather, consolidate with two-stage matching, contradiction detection, power-law decay (∼800 lines).
- `internal/embeddings/`—Embedding client with graceful degradation, vector store, training event recorder (∼400 lines).
- `internal/cortex/`—Seven cognitive primitives, knowledge graph, composite scoring, system prompt formatting (∼700 lines).

Database schema: 6 migrations adding `user_memories`, `dream_sessions`, embedding columns with HNSW indexes, `memory_training_events`, `memory_edges`, `contradicted_by` column, and encoding-time `salience` column.

Total: ∼3,200 lines of Go across 35+ files. The entire memory system runs on PostgreSQL with zero external dependencies.

8 Future Work

Six experimental directions are being developed in parallel:

1. **Context velocity decay**—Scale the forgetting rate by how fast the user's world is changing, not wall-clock time. High-activity users dream deeper; low-activity users dream lighter.
2. **Implicit negative training signals**—Track which search results the agent returns but does not use. The strongest signal for the scoring model is not what was retrieved—it is what was retrieved and ignored.
3. **NLI contradiction classifier**—Replace the LLM contradiction call with a 22M-parameter NLI model achieving 95% accuracy at near-zero cost. Fine-tune on Eva's own contradiction pairs.
4. **Stochastic memory replay**—A second consolidation pass on the dream session's own outputs, looking for meta-patterns: patterns that emerge from patterns. Inspired by the Phasor Agents finding that REM-like replay with perturbation improves performance by 45%.
5. **Multi-graph memory decomposition**—Stratify the knowledge graph into orthogonal semantic, temporal, causal, and entity layers following MAGMA's architecture [MAGMA, 2026]. Enable query-adaptive graph traversal in the RECALL scoring function, addressing the causally grounded retrieval gap identified by Du et al. [2026].
6. **MemoryAgentBench evaluation**—Benchmark the dreaming pipeline against the ICLR 2026 MemoryAgentBench suite [MemoryAgentBench, 2026], specifically the selective forgetting competency. Establish quantitative baselines for decay accuracy and contradiction resolution against the field.

9 Limitations

This work has several gaps that should be stated directly. There are no benchmark results. The architecture is described and deployed but not evaluated against standard benchmarks like LoCoMo, LongMemEval, or MemoryAgentBench. Until that evaluation is complete, the claims are architectural, not empirical.

The consolidation pipeline is LLM-dependent end to end. Extraction, semantic matching, and contradiction detection all route through a language model. If the LLM misclassifies a reinforcement as a contradiction—or misses a pattern entirely—the error compounds across dream sessions with no self-correction mechanism. The planned transition to a learned scoring model addresses one layer of this dependency, but the extraction and matching phases remain vulnerable.

The decay parameters have not been ablated. The power-law formula, the 0.15 base factor, the +0.05 reinforcement increment, the 0.1 expiry threshold—these were set by informed intuition, not experimental optimization. They produce reasonable behavior in production, but "reasonable" is not "optimal" and we have not yet demonstrated sensitivity to these choices.

Production deployment is early-stage. The system runs, dreams complete, memories consolidate and decay. But the user base is small and the total number of completed dream sessions is in the hundreds, not thousands. The architectural patterns hold under current load; whether they hold under 10× or 100× remains to be demonstrated.

Finally, the biological analogy is structural, not mechanistic. Synaptic homeostasis operates on continuous neural weights through biochemical cascades; our system operates on discrete confidence scores in PostgreSQL rows through scheduled batch jobs. The dynamics are isomorphic in their mathematical form—multiplicative downscaling with selective preservation—but the implementation substrates are fundamentally different. The analogy guided the design. It should not be mistaken for equivalence.

10 Conclusion

Kanerva [1988] demonstrated that high-dimensional distributed representations achieve robust storage and retrieval that degrades gracefully with noise. Graves et al. [2014] showed that differentiable external memory enables neural networks to learn algorithmic procedures. Packer et al. [2023] proposed treating LLMs as operating systems with tiered memory management. FadeMem [Wang et al., 2026] showed that biologically-inspired forgetting improves agent memory efficiency. Phasor Agents [Phasor Agents, 2026] proved that explicit wake/sleep separation expands stable learning by 67%. SleepGate [SleepGate, 2026] reduced proactive interference from $O(n)$ to $O(\log n)$ through sleep-cycle consolidation at the attention layer. AgeMem [Xu et al., 2026] and Memory-R1 [Yan et al., 2025] demonstrated that learned memory policies outperform hand-tuned heuristics. CLS-M [CLS-M, 2026] independently converged on the same biological blueprint—dual-store, consolidation, decay—achieving 96.5% context token reduction. MemoryAgentBench [MemoryAgentBench, 2026] confirmed that selective forgetting remains an unsolved competency across all current approaches.

What we have built sits at the intersection of these ideas, grounded in the biological mechanisms that have been solving the memory problem for several hundred million years. The hippocampus-neocortex transfer, mediated by sleep-dependent consolidation with reinforcement-driven selection pressure, is an architecture that works. The field is converging on this conclusion from multiple directions simultaneously. We translated it into seven primitives, implemented it in Go, and deployed it to production.

The dreaming pipeline runs. Contradiction detection is active. Power-law decay preserves what matters and forgets what does not. Training signals accumulate. The model will learn what matters from what actually mattered.

Synaptic homeostasis is multiplicative downscaling with selective preservation of potentiated connections. Applied to a vector of memory confidence scores with reinforcement-gated preservation and power-law trace-strength modulation, the dynamics are identical. The answer to AI agent memory is surprisingly biological. Do not build a bigger context window. Do not summarize harder. Let your AI sleep. Let it dream. Let it forget the noise and keep the signal.

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